

Diodos: Lección 26 julio

Ejemplo 3.2

Característica i-v

José Ramos López
Escuela de Ingeniería Eléctrica
Universidad de El Salvador
ELC-115

Ejemplo 3.2: circuitos con dos diodos

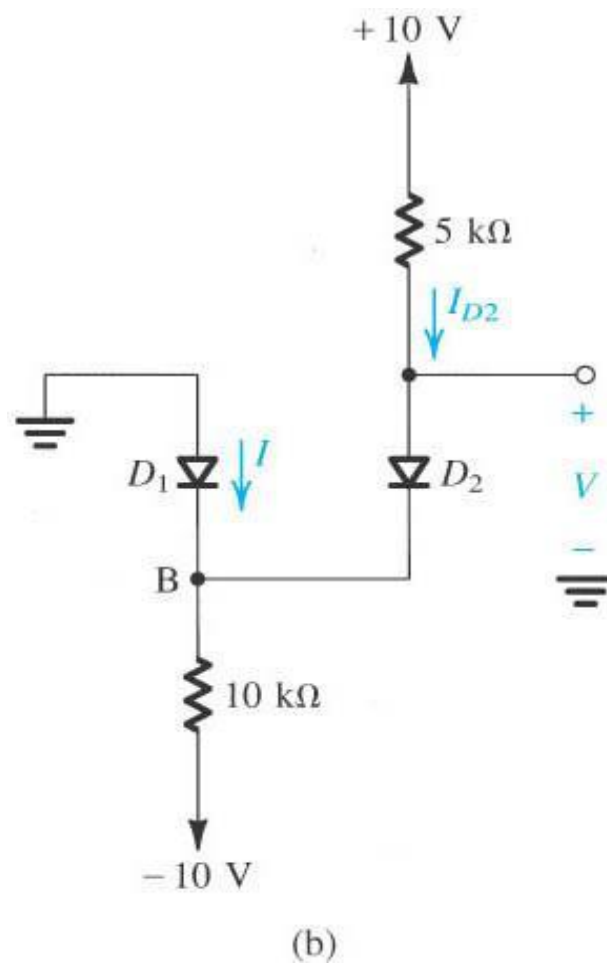
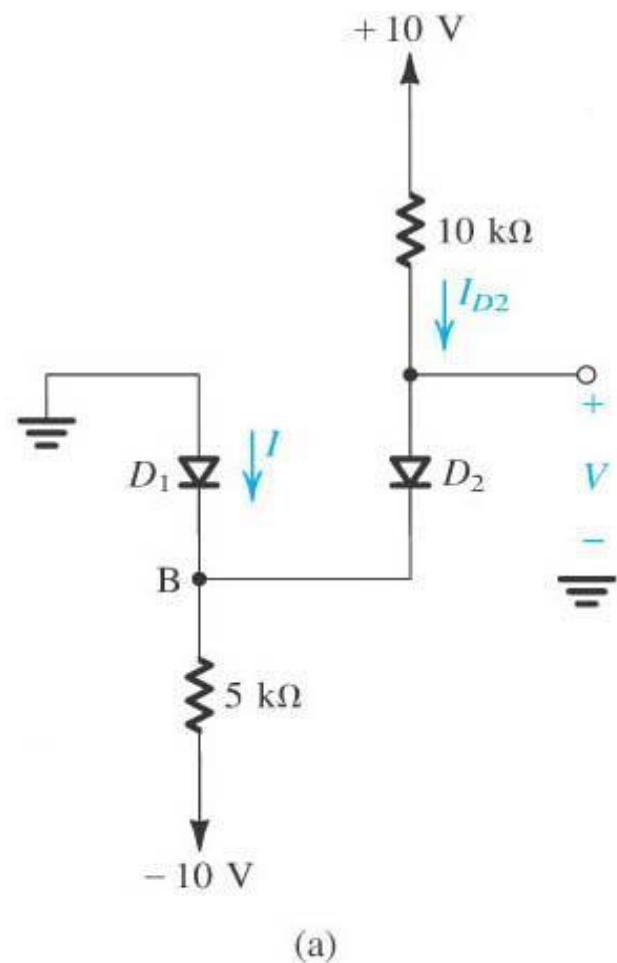
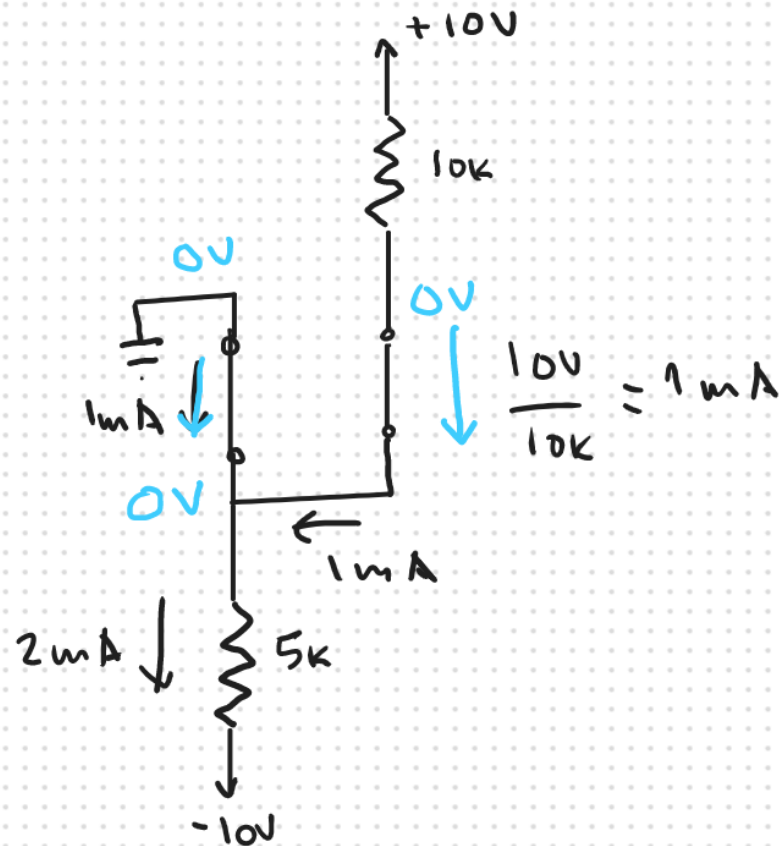
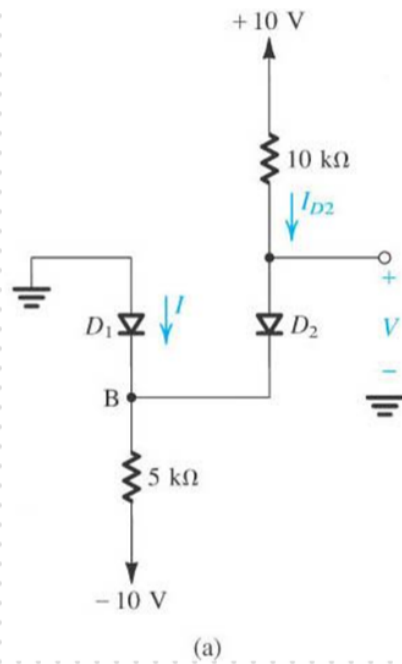
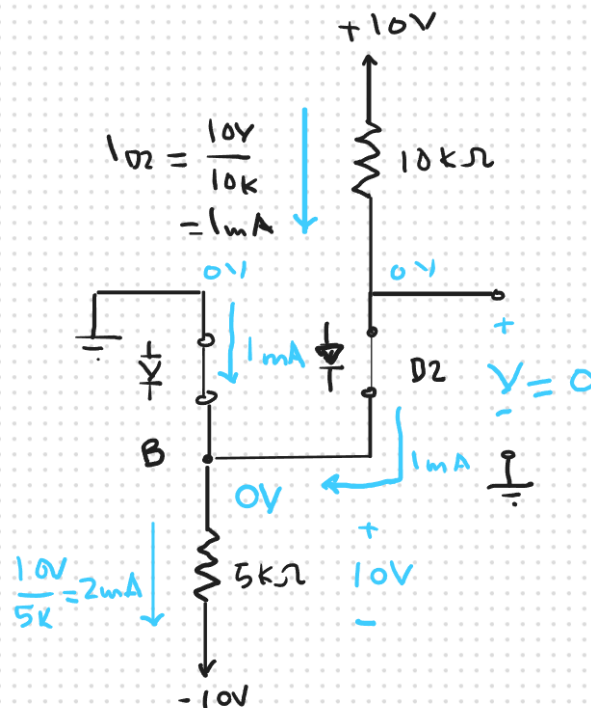
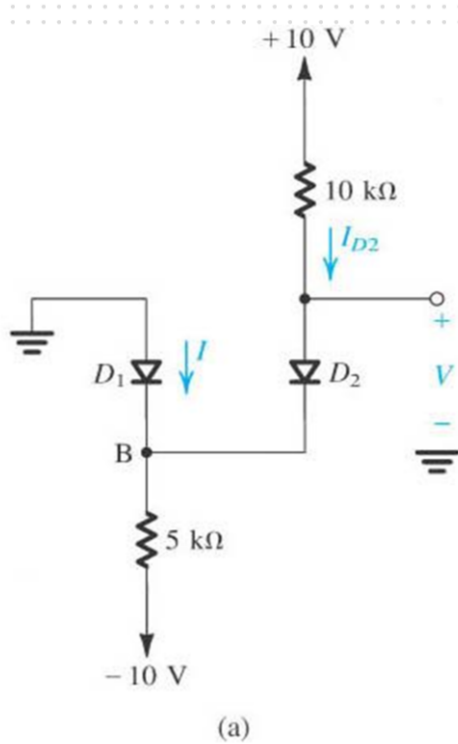
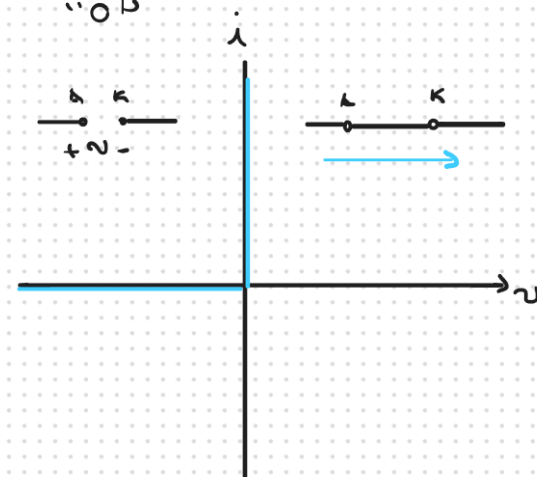


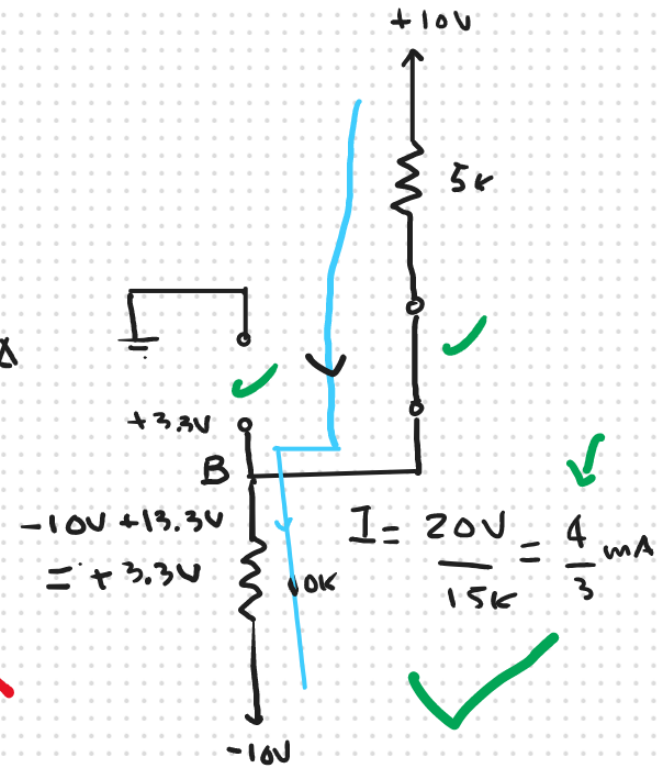
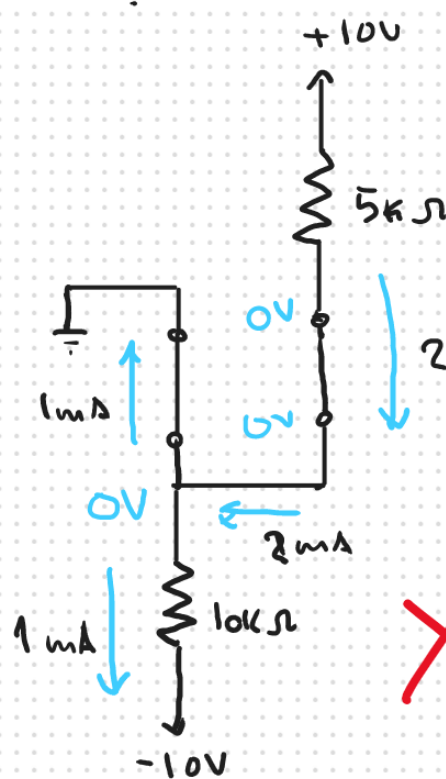
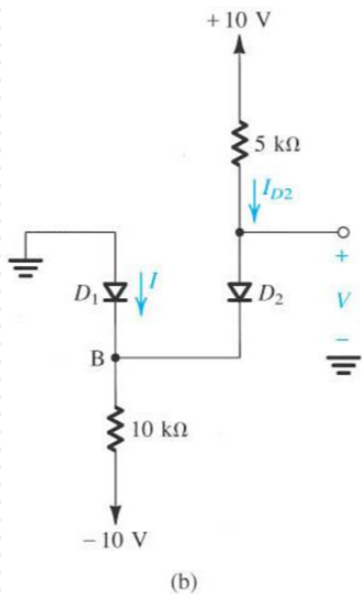
Figura 3.6 Circuitos para ejemplo 3.2.



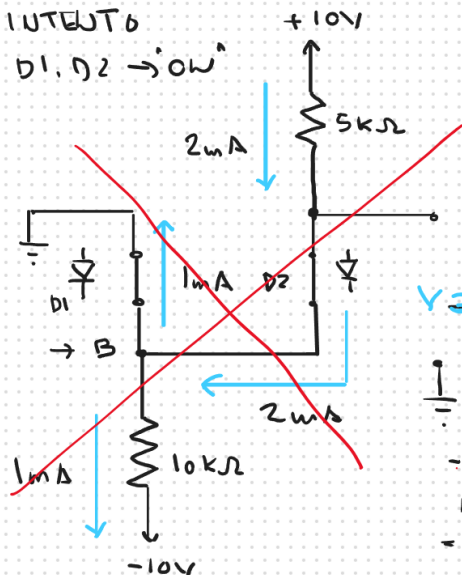


D1, D2
EN CONDUCCIÓN
"ON"

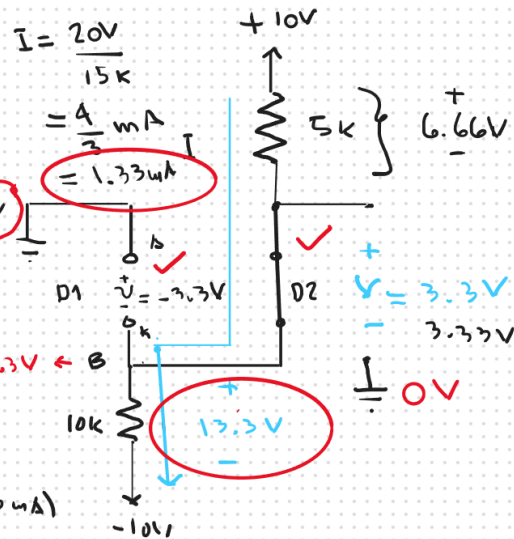




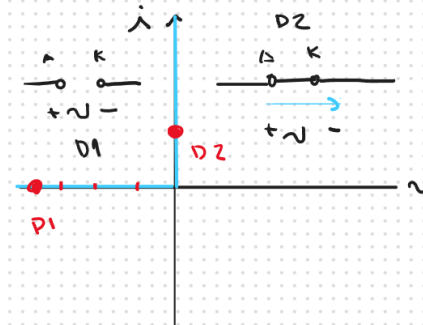
1er
INTENTO
D1, D2 $\rightarrow$ ON



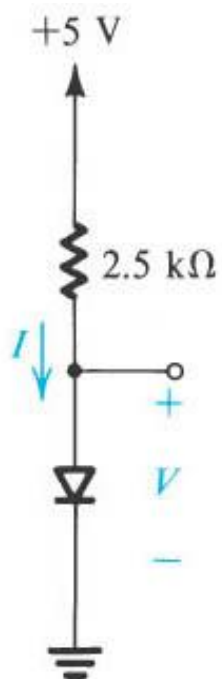
2º INTENTO: D1 $\rightarrow$ OFF; D2 $\rightarrow$ ON



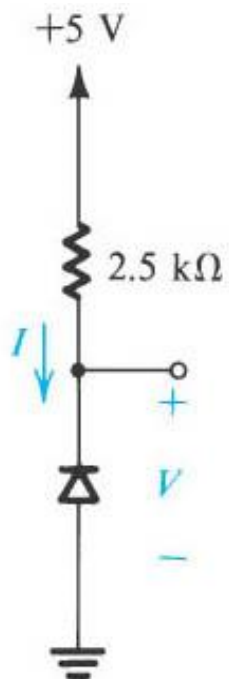
D1	D2
ON	ON
ON	OFF
OFF	ON
OFF	OFF



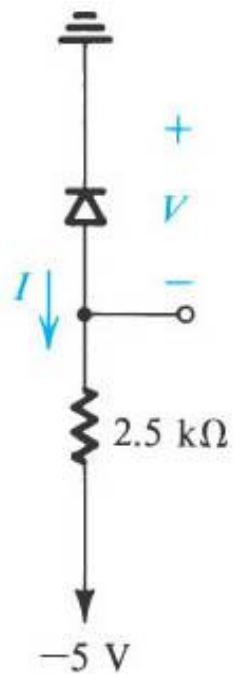
Ejemplo 3.4: circuitos varios



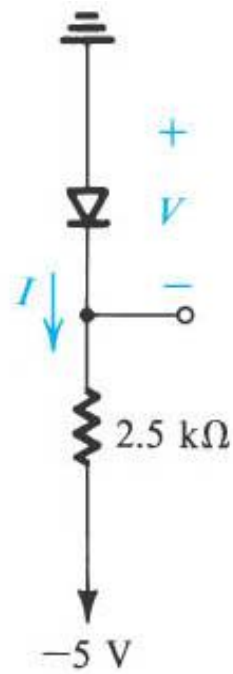
(a)



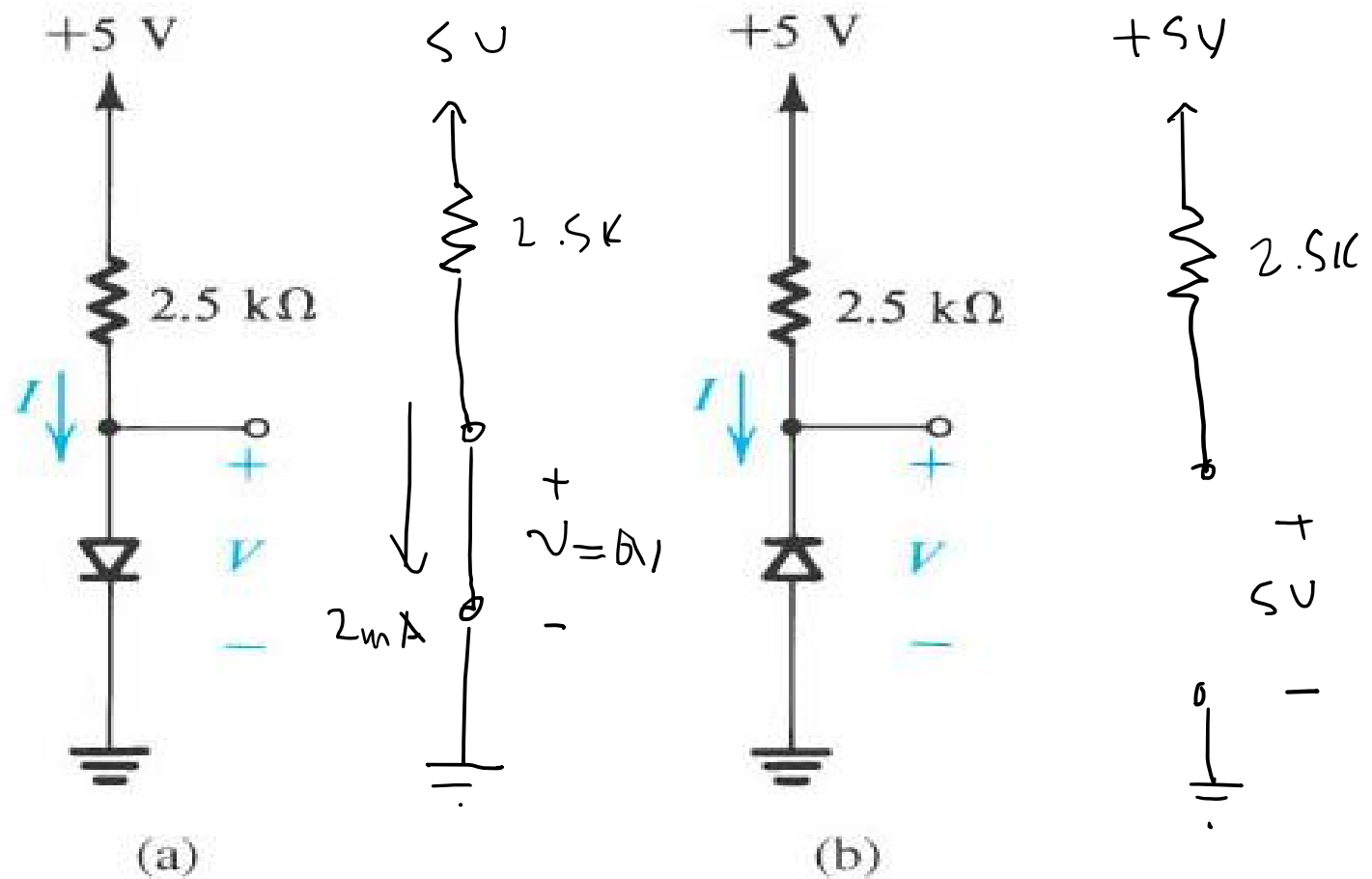
(b)

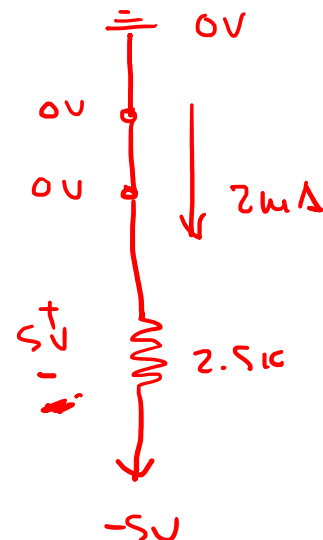
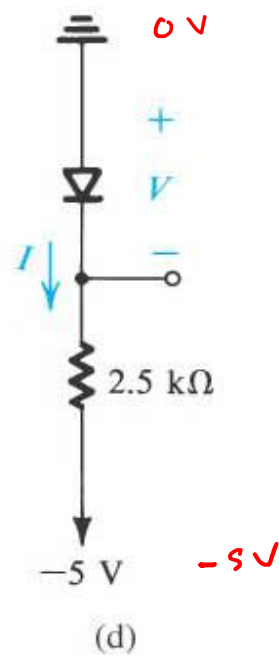
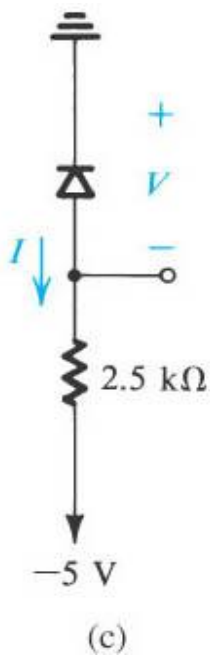


(c)

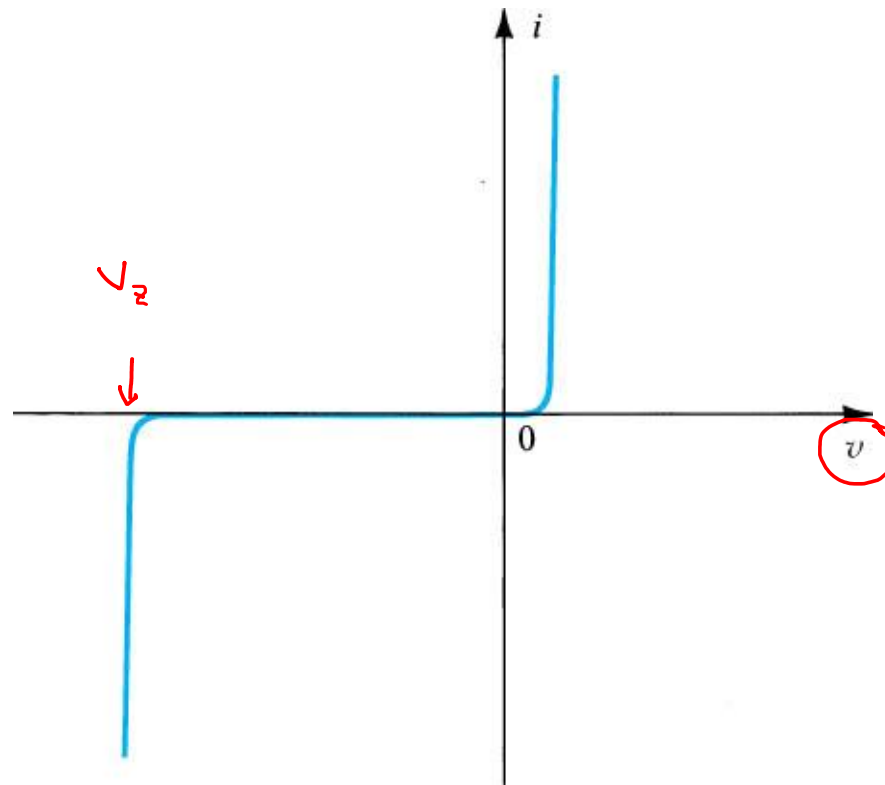


(d)

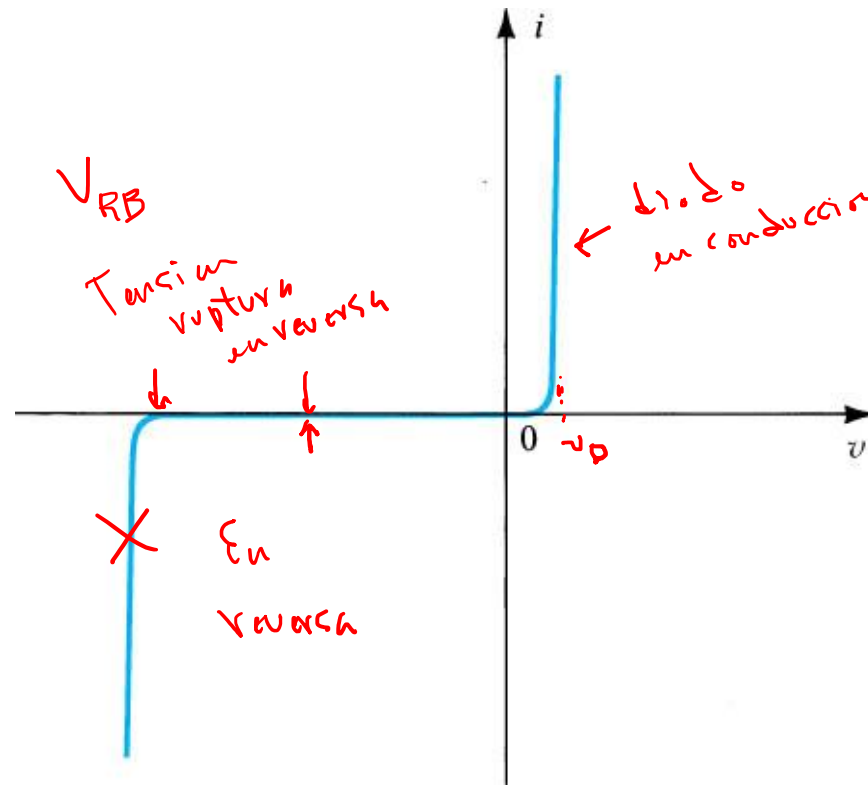




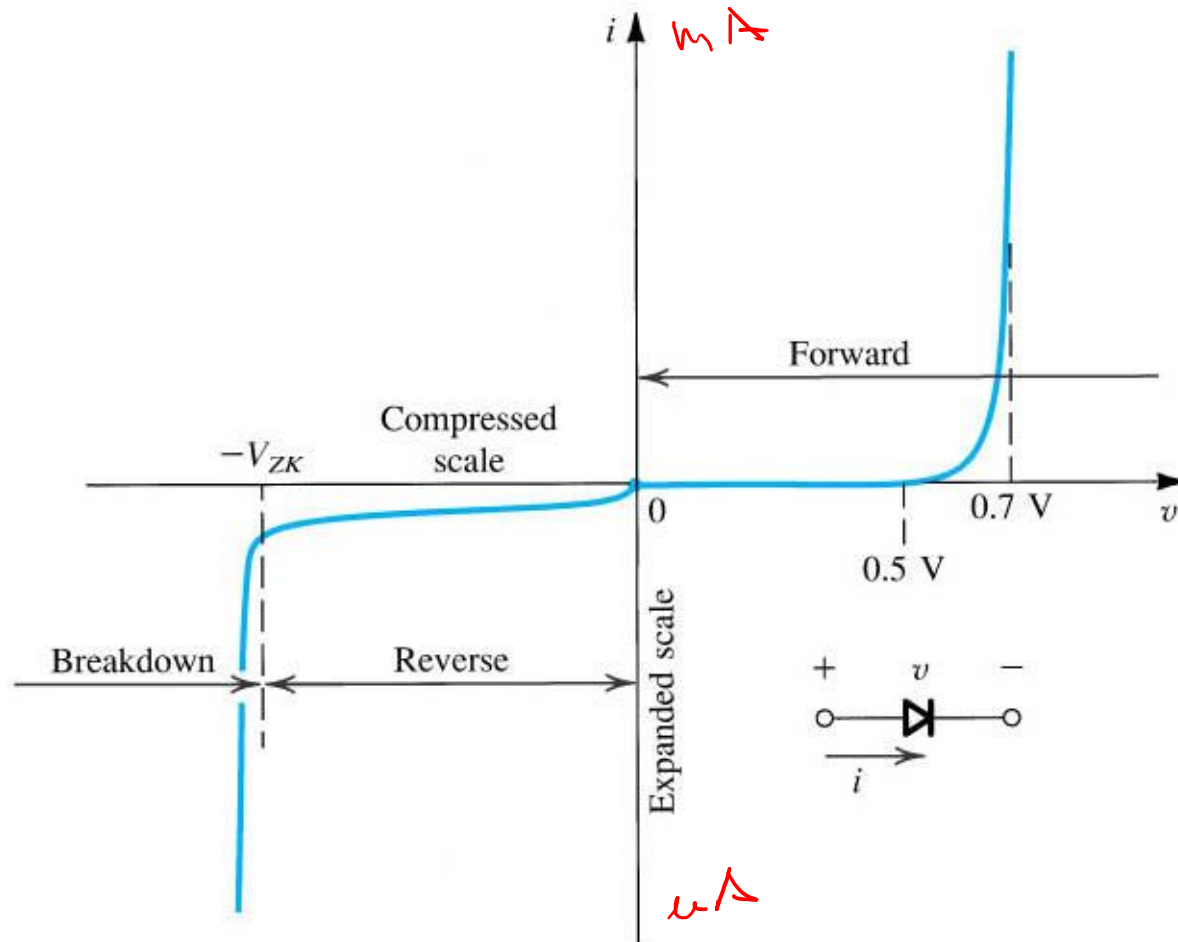
Característica i-v de diodo de unión



Características I-V de Diodos de Unión



Características I-V de Diodos de Unión



Relación I-V de un diodo. Se han ampliado o reducido algunas escalas para mostrar más detalles.

$$i = I_s \left(e^{\frac{v}{\eta VT}} - 1 \right)$$

$$V_T = \frac{kT}{q} = 25mV (@ 20^\circ C) = 25.8mV (@ 25^\circ C)$$

$$i \gg I_s$$

$$i = I_s e^{\frac{v}{\eta VT}}$$

$$v = \eta VT \ln \left(\frac{i}{I_s} \right)$$

Expresando en forma logarítmica

$$I_1 = I_s e^{\frac{v_1}{\eta VT}}$$

$$I_2 = I_s e^{\frac{v_2}{\eta VT}}$$

$$\frac{I_2}{I_1} = e^{\frac{v_2 - v_1}{\eta VT}}$$

$$v_2 - v_1 = \eta VT \ln \left(\frac{I_2}{I_1} \right)$$

$$v_2 - v_1 = 2.3 \eta VT \log \left(\frac{I_2}{I_1} \right)$$

$$i = I_s \left(e^{\frac{v}{\eta VT}} - 1 \right)$$

$$i \approx I_s e^{\frac{v}{\eta VT}}$$

I_s también es función de T

$$i_D = I_s \left(e^{\frac{v}{nV_T}} - 1 \right)$$

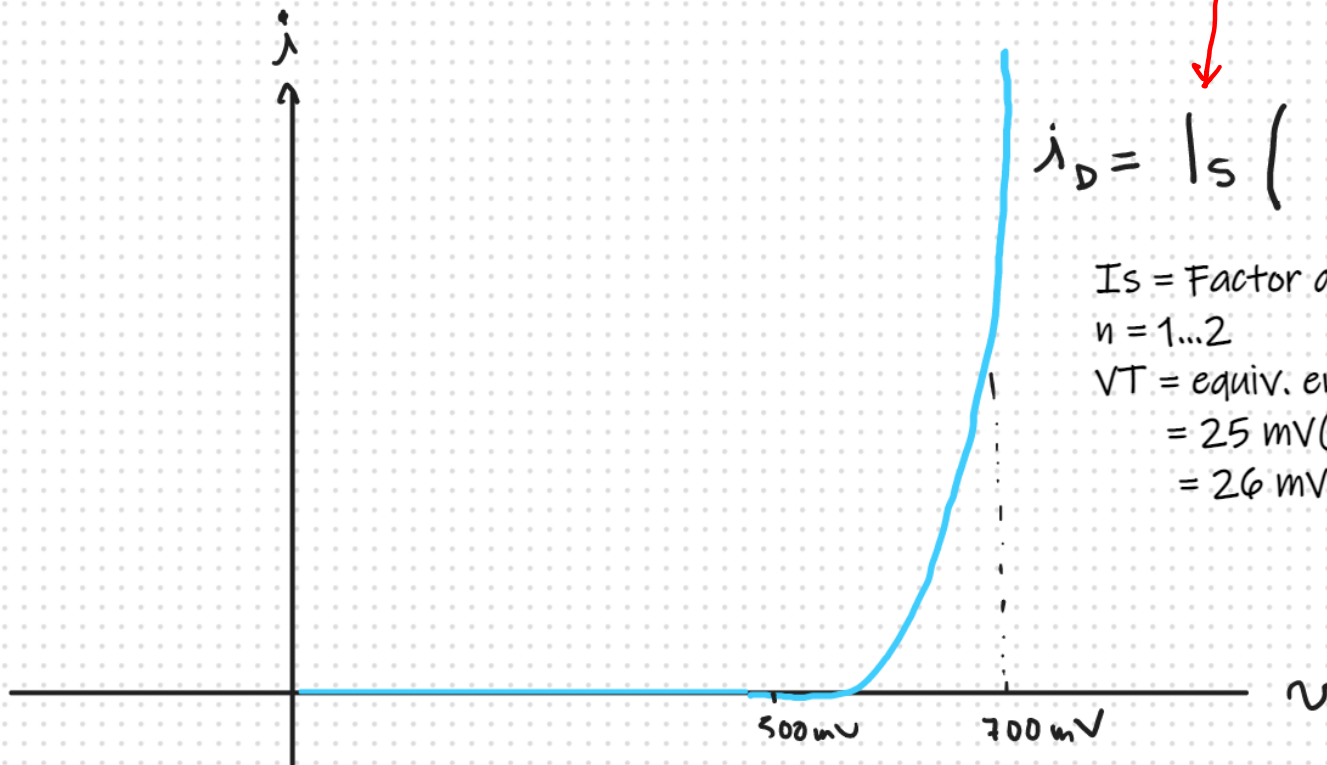
I_s = Factor de escala

$n = 1 \dots 2$

V_T = equiv. en volts de T

= 25 mV ($\approx 20^\circ\text{C}$)

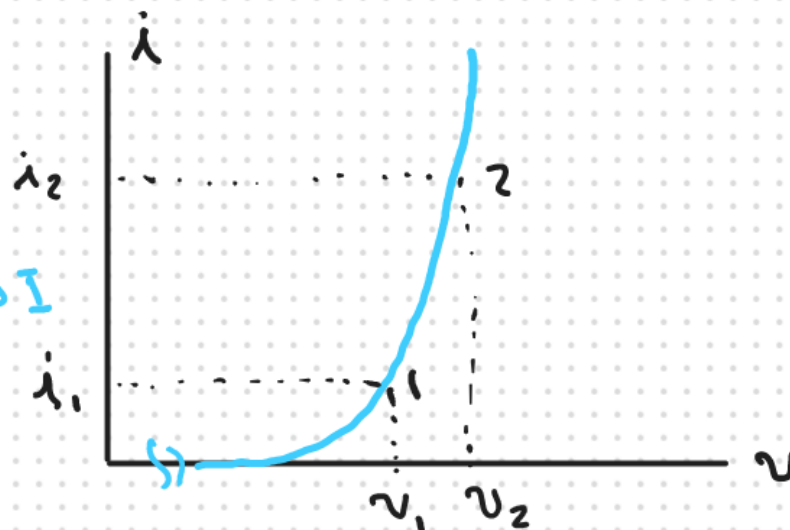
= 26 mV ($\approx 25^\circ\text{C}$)



DIODOS SILICIO

$$I_D \approx I_S e^{\frac{V}{nV_T}}$$

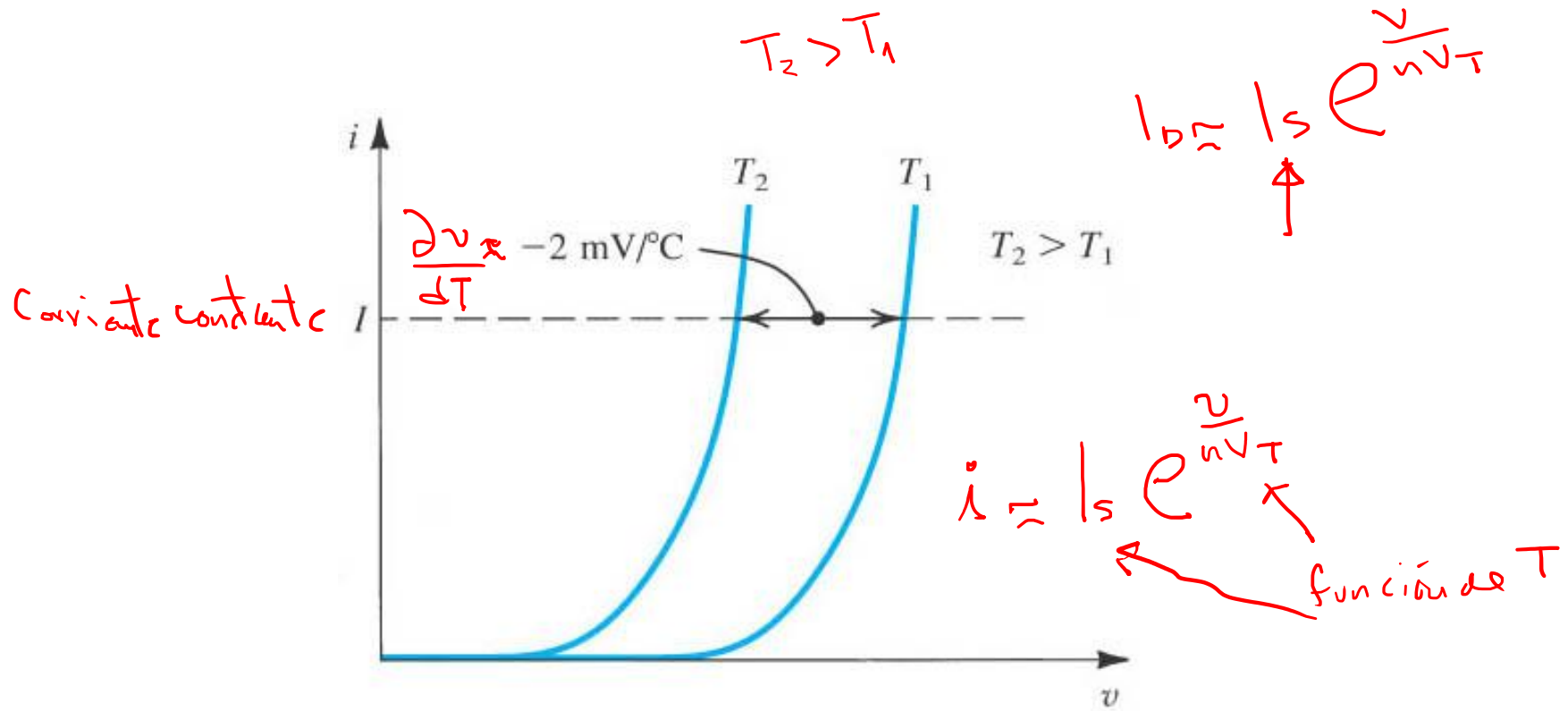
$$\left\{ \begin{array}{l} I_1 = I_S e^{\frac{V_1}{nV_T}} \\ I_2 = I_S e^{\frac{V_2}{nV_T}} \end{array} \right. \quad |$$



$$\left\{ \frac{I_2}{I_1} = \frac{e^{\frac{V_2}{nV_T}}}{e^{\frac{V_1}{nV_T}}} = e^{\frac{V_2 - V_1}{nV_T}} \right.$$

$$\Rightarrow V_2 - V_1 = nV_T \ln \left(\frac{I_2}{I_1} \right)$$

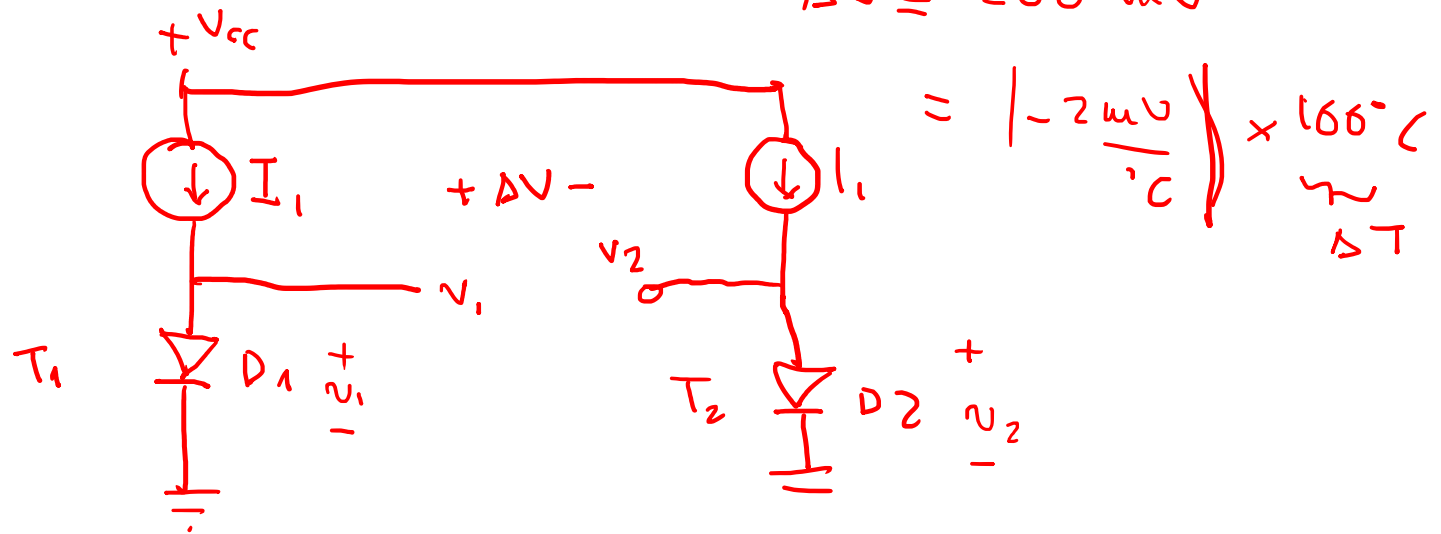
$$\Delta V \approx V_2 - V_1 = 2.3 nV_T \log \left(\frac{I_2}{I_1} \right)$$



Ilustrando la dependencia del diodo con la temperatura. A corriente constante, la caída de voltaje disminuye aprox. 2 mV por 1°C de incremento en temperatura.

ei $T_2 = T_1 + 100^\circ\text{C}$

$\Delta V = 200 \text{ mV}$



$= \left| -2 \frac{\text{mV}}{^\circ\text{C}} \right| \times 100^\circ\text{C}$

$\frac{dV}{dT} = -2 \frac{\text{mV}}{^\circ\text{C}}$

